Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Continuous Assessment Test (CAT) – I AUGUST 2026

Programme	:	M.TECH	Semester	:	01
Course Code & Course Title	:	MACSE517 & Computer Architecture and Organization	Class Number	:	CH2026270103467
Faculty	:	Dr. Christopher Columbus C	Slot	:	A1+TA1
Duration	:	90 mins	Max. Mark		50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1		A processor executes instructions using a 5-stage pipeline. Two candidate processor designs, Design A and Design B, have the following stage latencies (in picoseconds): **Design** / **Fetch** / **Decode** / **Execute** / **Memory** / **Write-back** A / 300 / 400 / 350 / 550 / 100 B / 200 / 150 / 100 / 190 / 140 Each pipeline stage requires an additional 20 ps to account for the pipeline register inserted between stages. Answer the following for both Design A and Design B: (a) Calculate: Clock cycle time, Latency of a single instruction, and instruction throughput for both the non-pipelined and pipelined implementations. **(3 Marks)** (b) You are allowed to split one pipeline stage into two equal-latency halves to reduce the clock cycle time. **(4 Marks)** (i) For each design, identify the pipeline stage to be split into two equal-latency sub-stages and justify your selection. (ii) Determine the new clock cycle time, latency, and throughput for the refined 6-stage pipeline. (iii) Compare the performance of the refined pipeline with the original 5-stage pipeline, and discuss whether the additional pipeline stage results in a significant performance improvement or demonstrates diminishing returns. (c)Assume each processor executes a program consisting of 100 instructions. For both Design A and Design B: **(3 Marks)** (i) Calculate the total execution time using both the original 5-stage pipeline and the refined 6-stage pipeline. (ii) Compare the execution times and determine the execution time saved by the refined pipeline over the original 5-stage pipeline.	10	1	K3

2		A warehouse robot operates in a dynamic environment and frequently executes branch instructions (e.g., if an obstacle is detected, turn left; otherwise, move forward) to respond to sensor inputs in real time. The robot's processor uses a 4-stage instruction pipeline. (a) Draw the pipeline execution diagram for a branch instruction followed by the next sequential instruction. Clearly indicate the pipeline stage at which the branch target is resolved, identify the type of hazard that occurs, and determine the number of stall cycles introduced. **(3 Marks)** (b) Describe any five techniques used to reduce or eliminate control hazards in pipelined processors. Illustrate each technique with a suitable example. **(7 Marks)**	10	1	K4
3		SmartLogic Automation Pvt. Ltd. is developing an embedded controller for an industrial automation system that monitors sensors, controls actuators, and executes firmware stored in ROM. The controller uses memory-mapped I/O, where memory and peripheral devices share the same address space. The available memory chips are 128 × 8 RAM and 512 × 8 ROM. The system requires 256 × 16-bit RAM, 1024 × 16-bit ROM, and two interface units, each consisting of 256 × 8-bit registers. As a hardware design engineer, design the memory organization and address decoding for the embedded controller. (a) Determine the number of RAM chips, ROM chips, and interface circuits required to implement the system. **(2 Marks)** (b) Design the memory address map, showing the address ranges assigned to the RAM, ROM, and the two interface units. **(3 Marks)** (c) Draw the memory organization (chip layout) diagram, clearly indicating the address lines, chip-select signals, data bus connections, and address decoding logic. **(5 Marks)**	10	2	K6
4		(a) Explain the cache mapping technique that eliminates conflict misses. Draw the cache lookup diagram for a fully associative cache, showing the flow of a memory address from the CPU to the cache, tag comparison, hit/miss detection, and access to main memory on a cache miss. (5 Marks) (b) Falcon Microsystems observes the following memory performance issues: (i) L1 cache misses result in long delays because data must be fetched from main memory, and pending read requests are delayed by write-back operations. Suggest suitable techniques to reduce the cache miss penalty and briefly explain how each technique improves performance. (3 Marks) (ii) Even on a cache hit, access time increases because the Virtual Address must first be translated to a Physical Address by the TLB before cache access. Name and explain a technique that allows address translation and cache indexing to occur simultaneously, including how the address is divided and the role of the TLB. (2 Marks)	10	2	K2
		A digital system designer is developing a 4K × 1 memory chip for a microcontroller-based data logging system using a two-dimensional memory organization.	10	2	K4

		(a) Determine the total number of memory locations and the number of address lines required for the 4K × 1 memory chip. Assuming the memory cells are arranged as a square two-dimensional array, calculate the number of rows and columns in the memory matrix. **(4Marks)** (b) Draw neat block diagrams of the 4K × 1 memory organization using a two-dimensional memory array. For each organization, clearly show the row decoder, column decoder/multiplexer, memory cell array, sense/write circuitry, address lines, and the data flow during both read and write operations.**(6 Marks)**			

**********All the best **************